Tab. 2: Fitted models selected by AIC for each vital rate. x represents the size, a the age, and the random effects are noted *year*, *pop*, *year* : *pop* for population effect nested in year effect, and ε for residual variance.

Survival probability for seedling

$$\text{logit}(s_1(x)) = \begin{cases} year_0 - 1.66(x-0.5) + (3.99 + year_1)(x-0.5)^2 & \text{if } 0.5 \leq x \leq 1 \text{ cm} \\ year_0 + 0.166 + (2.33 + year_1)(x-1) - 2.3(x-1)^2 & \text{if } 1 \leq x \leq 1.5 \text{ cm} \\ year_0 + 0.756 + (0.0292 + year_1)(x-1.5) + 0.0502(x-1.5)^2 & \text{if } 1.5 \leq x \leq 15 \text{ cm} \end{cases}$$

$Var(year_0) = 1.582$; $Var(year_1) = 0.02833$; $Cov(year_0, year_1) = -0.04404271$

Survival probability for rosettes

$$\text{logit}(s_2(x)) = pop_0 + year_0 + spline_1(x) + spline_2(a)$$

$$spline_1(x) = \begin{cases} 1.01(x-0.5) - 0.0958(x-0.5)^2 & \text{if } x \leq 2 \\ 2.52 + 0.246(x-4.5) - 0.0117(x-4.5)^2 & \text{if } x \geq 4.5 \end{cases}$$

$$spline_2(a) = \begin{cases} (0.713 + year_1)(a-2) - 0.478(a-2)^2 + 0.0709(a-2)^3 & \text{if } a \leq 6.5 \\ (0.723 + year_1)(a-6.5) + 0.48(a-6.5)^2 - 0.952(a-6.5)^3 & \text{if } a \geq 6.5 \end{cases}$$

$Var(pop_0) = 0.076$; $Var(year_0) = 1.443$; $Var(year_1) = 0.011$; $Cov(year_0, year_1) = -0.014$

Flowering probability

$$logit(f(x,a)) = -10.21 + pop_0 + 226.93x - 76.57x^2 + 39.27x^3 + (133.48 + pop_1)a - 54.57a^2$$

$Var(pop_0) = 2.252$; $Var(pop_1) = 0.09228$; $Cov(pop_0, pop_1) = -0.2041362$

Plant growth

$$g(y,x,a) = \frac{1}{\sigma(x,a)\sqrt{2\pi}} \exp\left(-\frac{(x-\mu(x,a))^2}{2\sigma(x,a)^2}\right)$$

$$\mu(x,a) = pop_0 + year_0 + poly(x) + spline(a)$$

$$poly(x) = 6.8681 + (175.3943 + year_1)x - 27.1381x^2 - 14.0231x^3$$

$$spline(a) = \begin{cases} (-0.793 + year_2)(a-1) + 0.109(a-1)^2 & \text{if } a \leq 6.5 \\ -1.08 + (0.401 + year_2)(a-6.5) - 0.527(a-6.5)^2 & \text{if } a \geq 6.5 \end{cases}$$

$$\sigma(x,a) = \exp(1.01 + 0.55\log(x) - 0.12\log(a))$$

$Var(pop_0) = 0.331$; $Var(year_0) = 0.3225$; $Var(year_1) = 0.0362$; $Cov(year_0, year_1) = 0.00468$; $Var(year_2) = 0.0579$; $Cov(year_0, year_2) = -0.0108$; $Cov(year_1, year_2) = -0.00135$

Fecundity (number of capitulas)

$$\log(c(x,a)) = 2.32734 + year_0 + 0.06752x + year_1 a + \varepsilon$$

$Var(year_0) = 0.2127$; $Var(year_1) = 0.0238$; $Cov(year_0, year_1) = -0.00506226$; $Var(\varepsilon) = 0.282$

Seedling size distribution

$$\omega(y) \sim \text{Gamma}(\log(0.2873 + pop_0 + year_0 + pop : year_0 + \varepsilon))$$

$Var(year_0) = 0.0249$; $Var(pop_0) = 0.0248$; $Var(pop : year_0) = 0.0781$; $Var(\varepsilon) = 0.0213$

Establishment rate

$$\xi = 0.338 + \varepsilon$$

$Var(\varepsilon) = 0.132$